AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Final Examination
Fall Semester 2024/2025

Date: 4 /Jan./2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: 100
- This exam consists of 5 questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [20 Marks]

For each question, choose the most appropriate answer. Copy the answer to your answer booklet. Each question carries 2 marks.

1. In the following method header, which word can be capitalized without affecting the correctness of the code?

```
public static void calculateTax(double[] x)
```

   a. `public`
   b. `void`
   c. `calculateTax`
   d. None of the answers

2. Which of the following statements about the `continue` statement in Java is correct?
   a. It can be used to skip an iteration in a loop and continue with the next iteration.
   b. It can only be used within for loops, not while or do-while loops.
   c. It terminates the loop entirely and transfers control to the next statement after the lo
   d. Both a and b are correct

3. The part of the program where a variable can be accessed is the variable's _____.
   a. scope
   b. declaration
   c. initialization
   d. constant

4. What does `s` reference in the Java statement:

```
String s = ("Happy".charAt(0)=='h'?"1":"2");
```

   a. `"1"`
   b. `'h'`
   c. `"2"`
   d. `'H'`

5. Which of the following is a Ternary operator?
   a. ++
   b. *
   c. (? :)
   d. /

6. In the below code, which line has error?

```
1 int[] a;
2 a={1,5,7,8};
3 System.out.print(a[1]);
```

   a. 1
   b. 2
   c. 3
   d. No error in the code

7. In the below code, which line has error?

```
1 for (int i=0; i<3; i++)
2   System.out.println(i);
3 System.out.println(i);
```

   a. 1
   b. 2
   c. 3
   d. No error in the code

8. In Java, we can declare a method inside the body of another method.
   a. True
   b. False

9. Benefits of methods include:
   a. Information hiding
   b. Reusability
   c. Reduce complexity
   d. All answers are correct

10. In the following array declaration, What is the value of `x[1][1]` ?

```
int [][] x = {{4,1,6,5}, {0,2,3,9}};
```

   a. 4
   b. 1
   c. 0
   d. 2

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Question 2 [20 Marks]

Trace the following pieces of code and write their exact output:

1. (4 Marks)

```
System.out.println(Math.abs(-9.5));
System.out.println(Math.floor(-9.5));
System.out.println(Math.round(9.5));
System.out.println((int)9.5);
```

2. (4 Marks)

```
for (int i=10;i<=100;i+=5){
    if (i==20)
        break;
    System.out.println(i);
}
```

3. (4 Marks)

```
int x = 100;
switch (x++){
   case 100: System.out.println("A"); break;
   case 101: System.out.println("A+"); break;
   default: System.out.println("NONE");
}
System.out.println(x);
```

4. (4 Marks)

```
int[] array = {7,4,2,9,8};
array[1] += 2;
array[2]++;
array[3] = array[0]+array[4];
array[4] = array[0];
for (int i = 0; i < 4; i++)
   System.out.println(array[i]);
```

5. (4 Marks)

```
public class FinalExam {
  public static void main(String[] args) {
      int x = 10;
      multiply(x);
      System.out.println(x);
      x++;
      multiply(x);
      System.out.println(x);
  }
  public static void multiply (int a) {
      a = a * 5;
      System.out.println(a);
  }
}
```


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Question 3 [15 Marks]

1. Modify the following code such that `while` is used instead of the `for`. What is the exact output of the code? (7 Marks):

```
int i=10;
while(i>2)
{
    i-=2;
    System.out.println(i);
    i-=2;
}
```

2. Complete the following code to calculate the average of the numbers in the first column in the initialized 2D array `table` (8 Marks).

```
int [][] table = {{12,7,11,22},{13,9,12,10},{15,11,13,13}};
int sum;

.....
.....
.....
.....
.....

System.out.printf("The average of the first column is:
%.2f",average);
```

For questions 4 and 5 (next page), copy the below code into your answer booklet and continue the rest of the code according to the given specifications.

```
import java.util.Scanner;
public class FinalExam {
      public static void main(String[] args) {
          Scanner input = new Scanner(System.in);

    }
}
```


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Question 4 [20 Marks]

Write Java code to do the following:

1. Prompt the user to enter the names of tools available in a workshop ("finish" to stop).
2. The program should count and print the number of "hammers" available in addition to the total number of tools entered.

Here is a sample of the program run to help you understand how the program works:

```
Enter the names of the available tools ("finish" to stop):
hammer chisel wrench drill hammer chisel hammer finish
The number of hammers available is: 3
The total number of tools available is: 7
```

Question 5 [25 Marks]

Write Java code to do the following:

1. Write a Java method called `average()` that takes an array of real numbers as an argument. The method calculates and returns the average of the array elements.
2. In the main method, create a one dimensional array of 12 real numbers.
3. Read the monthly expenses for 12 months and store them in the array you have created.
4. Call the method `average()` to calculate the average monthly expense. Print the result.
5. Find and print how many months had expenses above the average.

Here is a sample of the program run to help you understand how the program works:

```
Enter your monthly expenses in 12 months:
220.5 251.75 300.5 500.9 450.9 377.9 215 395.5 412.5 190.6 300 170.5
The average monthly expense is: 315.0
There are 5 months that had expenses above the average
```

End of question

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